

QJazz

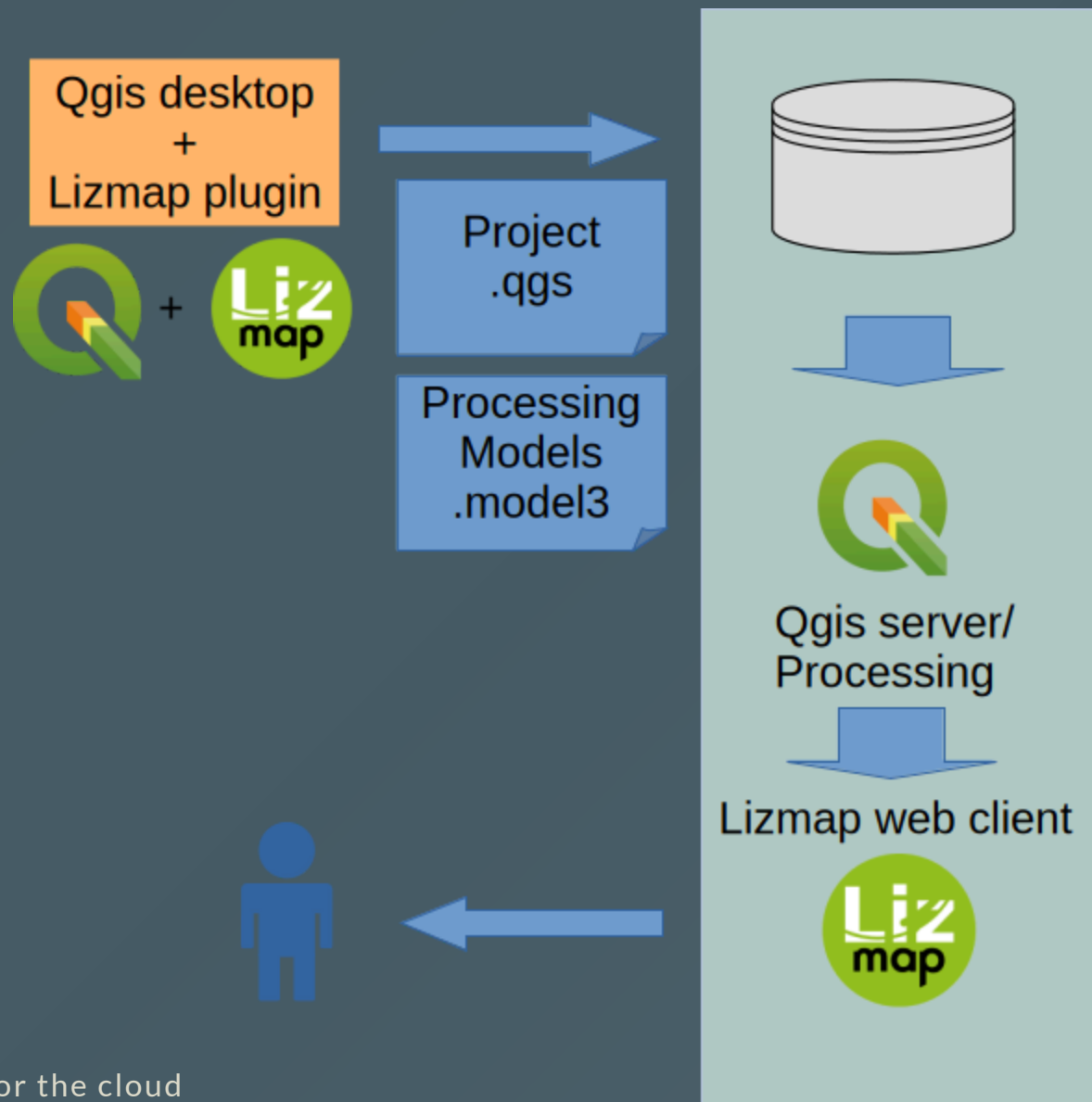


QGIS server solution for the cloud

Return of experience from GIS hosting services with Lizmap and QGIS server

<https://github.com/3liz/qjazz>

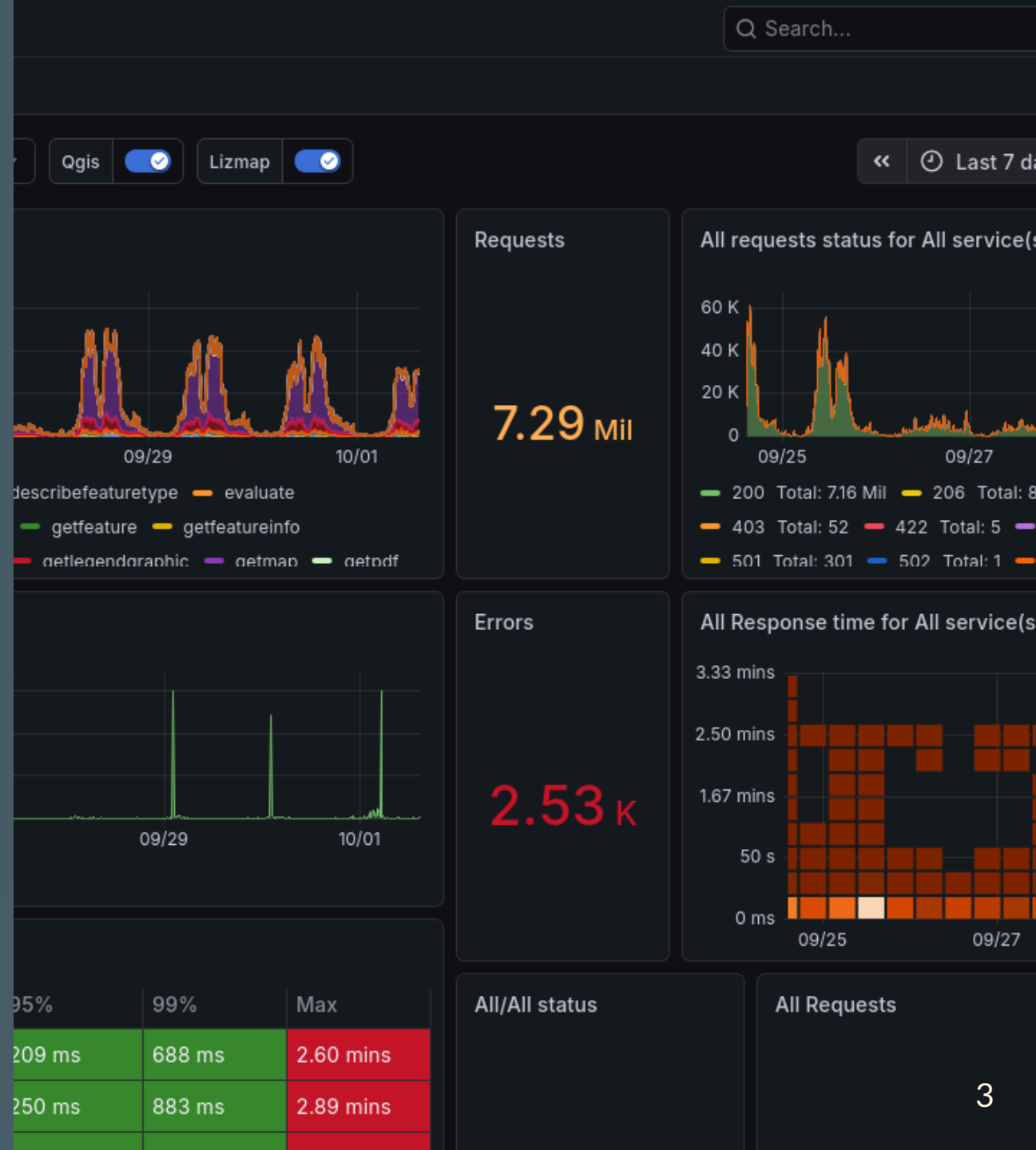
GIS hosting services with Lizmap and QGIS server



GIS hosting services with Lizmap and QGIS server

Numbers:

- 54 physical servers
- 529 PostgreSQL databases
- 472 lizmap instances
- 140 Services QGIS server ~ 1200 QGIS server workers
- 1870 QGIS projects



Problems to solve:

- Scalability
- Distributed architecture
- Monitoring
- Zeroconf (ideally !)
- Security

Dealing with issues

- Projects with many layers: up to 200 layers per project
- Loading times of several minutes
- Memory issues
- High latency requests (mainly remote services)
- Stuck server instances

When things go wrong



QJazz

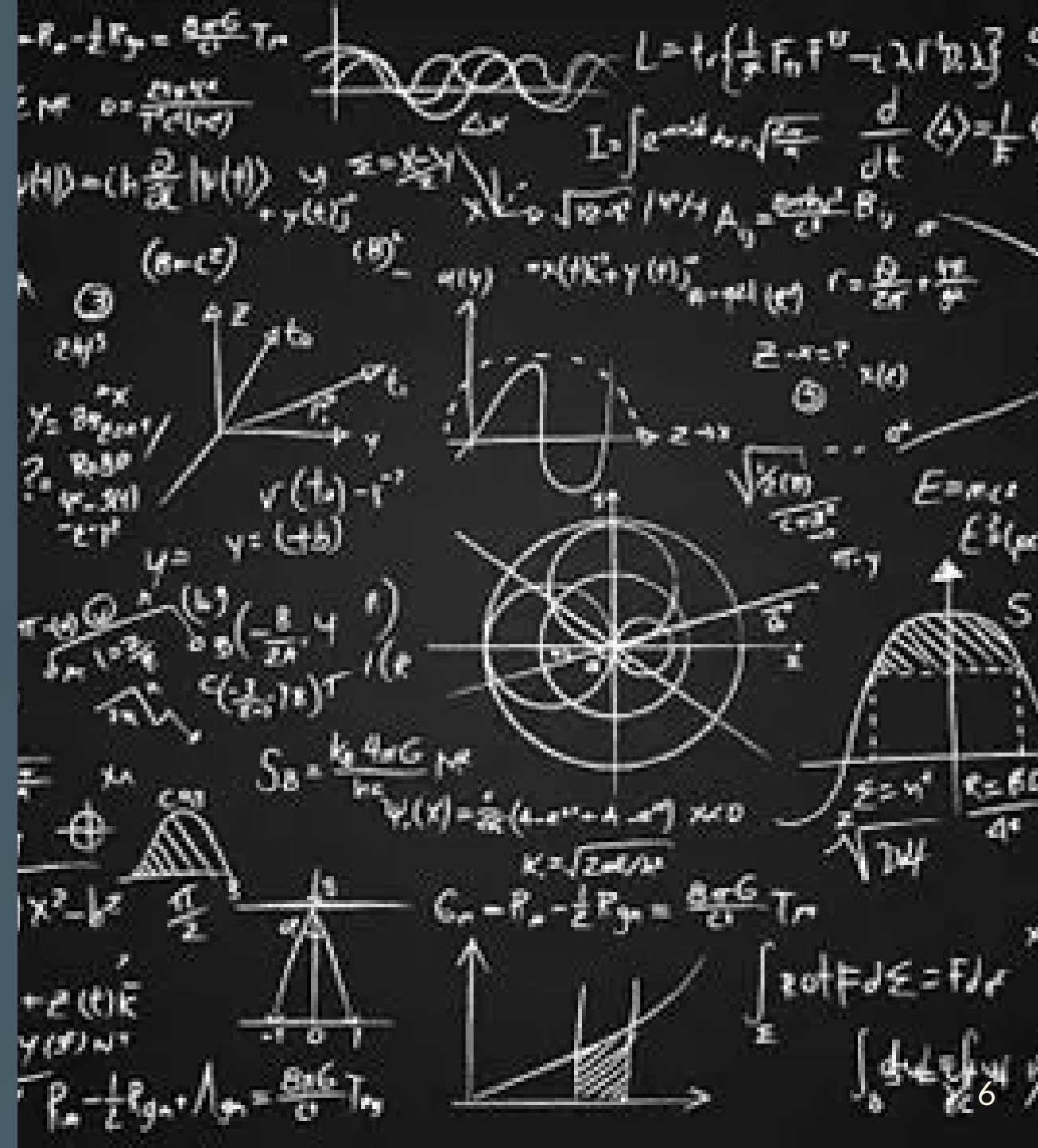
Microservices with gRPC protocol:

<https://github.com/3liz/qjazz>

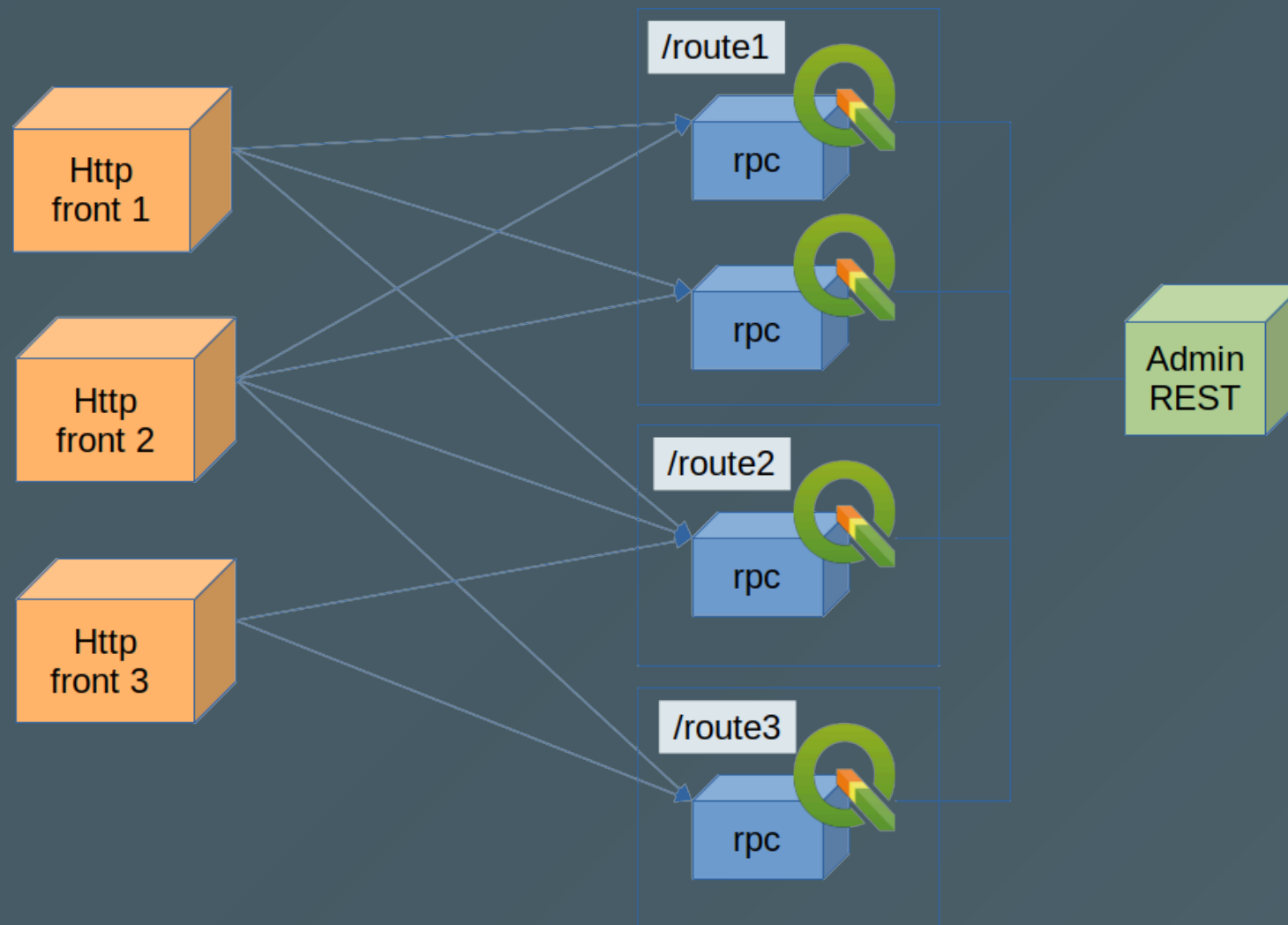
- Built-in Load balancing
- Built-in healthcheck support
- Dedicated administration service
- Bi-directional streaming support
- Independent from HTTP front-end



QGIS UC 2026 - QGIS Server ready for the cloud

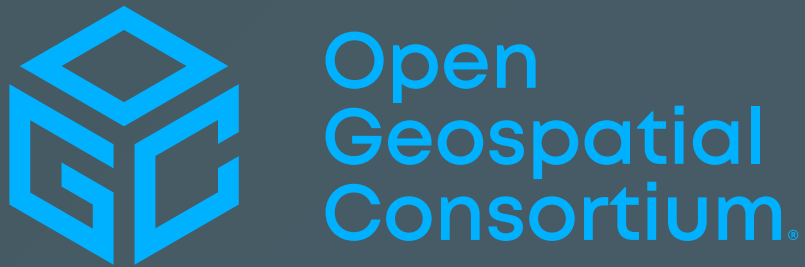


QJazz overview



QJazz services

- OGC OWS services (WMS, WFS, ...) - OGIS server native services
- STAC catalogs view of Projects and layers
- OGC API Maps - <https://ogcapi.ogc.org/maps/>
- OGC API Processes - QGIS Processing as a service
- Support for S3 backend (Projects and data)



QGIS Projects management in QJazz

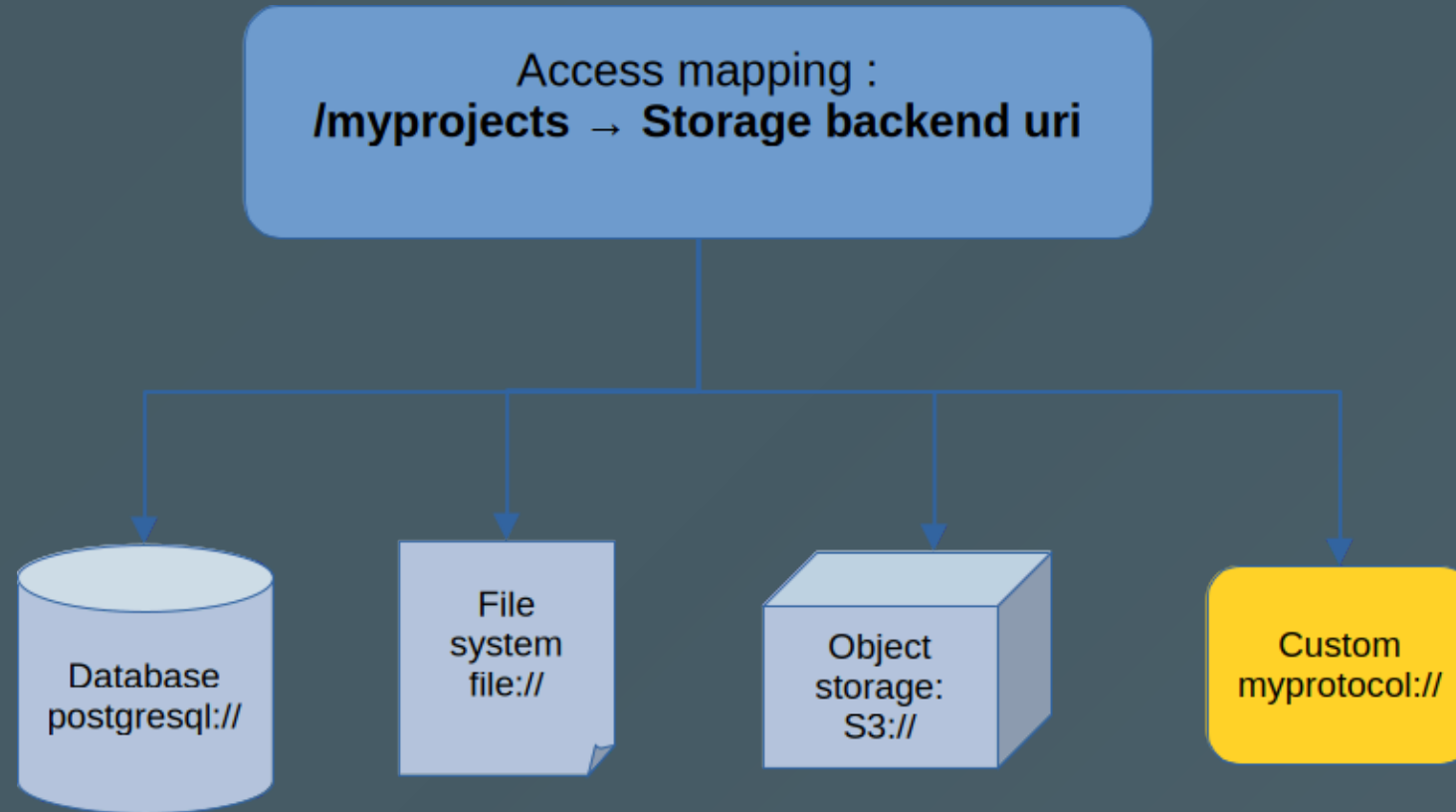
From a 'project as resource' perspective

- Consider Project as an application
- Corollary: QGIS server is an application server
- Control what is published (from a customer perspective)
- Keep some level of flexibility (dynamic caching)

From a deployment perspective

- Projects partitioning (routes)
- Handle access controls to backend Apis
- Easy to scale

Qjazz project's storage



Performances considerations

Horizontal scalability governance

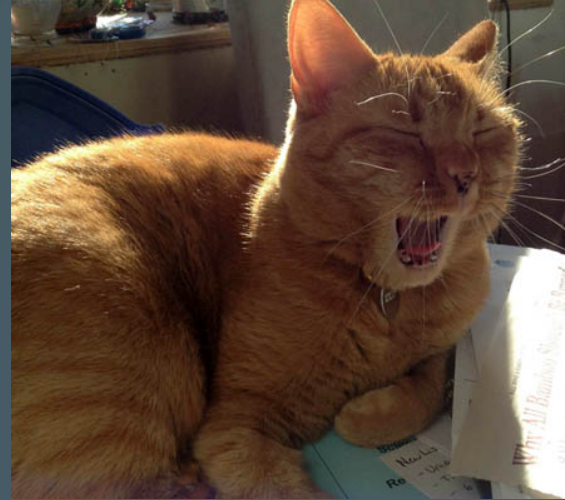
- What is the expected request rate ?
- How request distribute on projects ?
- How many different projects I have to handle ?

Impact on request processing

- How many layers in my project ?
- Data volumetry ?
- Accesses to database backends/external services ?

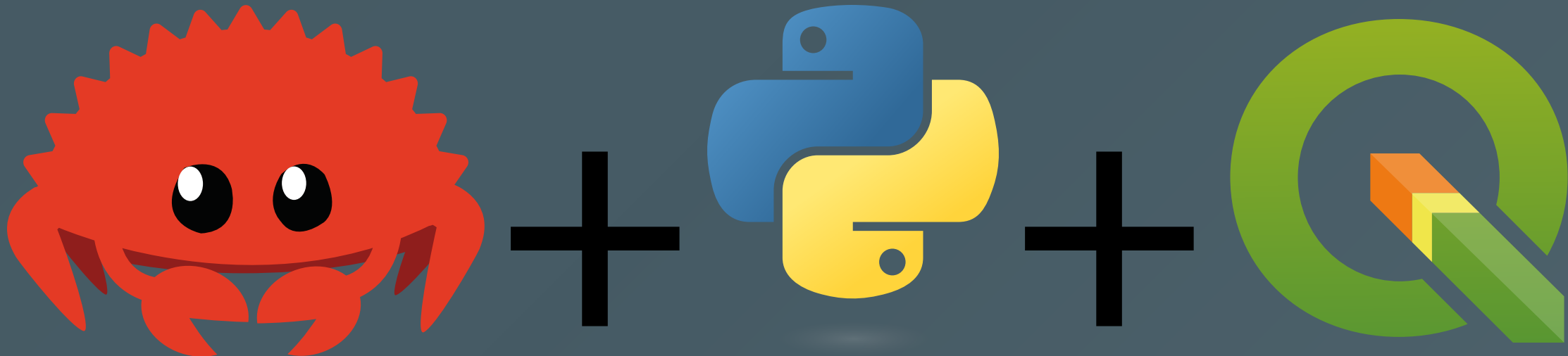


There is no single strategy to rule all your projects



Conclusion

- Revisiting QGIS server as an application server
- Cloud friendly integration (QGIS servers as micro-services)
- Address the problem of serving multiple use cases with many different kind of QGIS projects.



Thank you



Questions ?